Complete Time Complexity

O(1):-

```
n=100
i = 1
count=0
while i<=n:
    count=count+1
    if i==1:
        break
    i=i+1
print(count)

#O/P:-
if n=10     O/P = 1
if n=100    O/P = 1
if n=1000   O/P = 1
# i.e time complexcity is o(1)
```

O(n)

```
n=100
i = 1
count=0
while i<=n:
    count=count+1
    i=i+1
print(count)

#O/P:-
if n=10     O/P = 10
if n=100    O/P = 100
if n=1000   O/P = 1000
# i.e time complexcity is o(n)
```

O(n^2)

```
n = 10
count = 0
for i in range(n):
    for j in range(n):
```

```
        count += 1
print(count)
#O/P:-
if n=10     count = 100
if n=100    count = 10000
if n=1000   count = 1000000
# i.e time complexcity is o(n^2)
```

O(log(n))

```
n = 100
i = 1
count = 0

while i <= n:
    count += 1
    i = i * 2

print(count)

# O/P:-
7
# hear value of i is 1 → 2 → 4 → 8 → 16 → 32 → 64 → 128(terminated)

n = 10      count = 4
n = 100     count = 7
n = 1000    count = 10
```

Explaine:

---

i = 22222.....k-times
i = 2^k
loop is termination when i<=n
i.e
2^k = n
k = log2(n)

```
i = n = 10
count = 0

while i >= 1:
    count += 1
    i = i // 3
print(count)
```

```
# O/P:-
n = 10      count = 3
n = 100     count = 5
n = 1000    count = 7
```

i = n / 3 / 3 / 3 ... (k times)
i = n / 3^k
loop is termination when i<=1
1 = n / 3^k
i.e 3^k = n

k = log3(n)

O(nlog(n))

```
n = 10
count = 0

for i in range(n):          # O(n)
    j = 1
    while j <= n:           # O(log n)
        count += 1
        j = j * 2
print(count)
# O/P:-
n = 10      count = 40
n = 100     count = 700
n = 1000    count = 10000
```

`Outer loop runs` - n times

`Inner loop runs` -
i = 22222.....k-times
i = 2^k
loop is termination when i<=n
i.e
2^k = n
k = log2(n)

`Total = n × log n`
i.e final time complexcity is - **O(n log n)**